A paleomagnetic study of the lava flows within the Copper Harbor Conglomerate, Michigan: new results and implications

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New paleomagnetic data have been obtained from the interbedded lava flows within the Copper Harbor Conglomerate on Michigan's Keweenaw Peninsula. Previous paleomagnetic studies of these lava flows, known collectively as the Lake Shore Traps, have produced contradictory results. To investigate the cause of these conflicting results, 30 sites encompassing the most complete section of lava flows possible were collected and analyzed.

Well-defined characteristic directions of magnetization were isolated using either alternating-field or thermal demagnetization or a combination of both. These directions of magnetization are interpreted as primary magnetizations acquired during the original cooling of the lavas. Hysteresis, thermomagnetic, and petrographic analyses suggest the carrier of magnetization is a pseudo-single-domain, titanium-poor magnetite that has undergone some high-temperature oxidation.

Site means determined from the 30 lava flows define three distinct directional clusters. Each cluster of directions corresponds to a different stratigraphic package of lava flows with the Copper Harbor Conglomerate. Between-site dispersion for each stratigraphic package or unit is much less than the expected value for Keweenawan-age rocks. Therefore, we suggest that most of the lava flows in each unit were extruded rapidly and that within an individual stratigraphic unit, paleosecular variation has not been adequately sampled. This explains why previous studies on the Lake Shore Traps have produced such different results; each study did not sample the entire range of directions possible in these lava flows.

The paleomagnetic pole calculated from the 30 site-mean virtual geomagnetic poles is located at 22.2°N, 180.8°E (k = 35.0; A_{95} = 6.5°). The new Copper Harbor pole is now located in the appropriate chronological position with respect to the underlying Portage Lake Volcanics and the overlying Nonesuch Shale on the Keweenawan apparent polar wander path. The similarity of our Copper Harbor pole to that of the Portage Lake Volcanics reinforces the idea that the Copper Harbor Conglomerate is more closely related in time to the Portage Lake Volcanics than to the Nonesuch Shale.

De nouveaux résultats paléomagnétiques ont été obtenus pour les coulées de lave interstratifiées dans le Conglomérat de Copper Harbor, de la péninsule du Michigan keweenawienne. Les études paléomagnétiques antérieures de ces coulées de lave, collectivement appelées les Trapps de Lake Shore, ont fourni des résultats contradictoires. Afin de rechercher les raisons qui ont conduit à ces résultats incompatibles, 30 sites représentant le plus complètement possible la coupe des coulées de lave ont été échantillonnés et analysés.

Les directions d'aimantation bien définies ont été isolées en utilisant soit la désaimantation par champ alternatif ou soit par chauffage ou une combinaison des deux méthodes. Ces directions d'aimantation sont interprétées comme étant des aimantations primaires acquises durant le refroidissement original des laves. Les analyses d'hystérésis, thermomagnétique et pétrographique suggèrent que l'aimantation est portée par un pseudo-domaine de magnétite pauvre en titane ayant subi une oxydation à température élevée.

Les moyennes par site étudié, calculées pour les 30 coulées de lave, définissent trois groupes directionnels. Chaque groupe de directions correspond à un ensemble stratigraphique de coulées de lave qui est distinct du Conglomérat de Copper Harbor. La dispersion entre les sites pour chaque ensemble ou unité stratigraphique est inférieure à la valeur anticipée pour les roches d'âge keweenawien. Par conséquent, nous considérons que la majorité des coulées de lave dans chaque unité sont le produit d'extrusions rapides, et qu'à l'intérieur d'une unité stratigraphique individuelle, l'échantillonnage a été inadéquat et ne permet pas l'évaluation de la variation paléoséculaire. Ceci explique pourquoi les études réalisées auparavant sur les Trapps de Lake Shore ont fourni de tels résultats contradictoires; dans chaque étude l'échantillonnage n'a pas couvert la totalité du champ des directions possibles dans ces coulées de lave.

Le pôle paléomagnétique calculé à partir des moyennes des 30 sites des pôles géomagnétiques virtuels est localisé à 22,2°N, 180,8°E (k = 35,0, A_{95} = 6,5°). Cette nouvelle localisation du pôle de Copper Harbor occupe maintenant une position chronologique appropriée relativement aux Volcanites de Portage Lake sous-jacentes et au Shale de Nonesuch sur la courbe de la dérive apparente des pôles pour le Keweenawien. Cette similitude de la position des pôles de notre unité de Copper Harbor et des Volcanites de Portage Lake renforte l'idée que le Conglomérat de Copper Harbor est plus proche chronologiquement des Volcanites de Portage Lake que du Shale de Nonesuch.

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Introduction

The thin interbedded lava flows within the Copper Harbor Conglomerate have been the focus of several paleomagnetic studies over the last 30 years. The most recent study by Halls and Palmer (1981) resulted in a paleomagnetic pole position of 34.9°N, 175.6°E (precision parameter k = 239, semi-axes dp and dm of the 95% ellipse of confidence about the paleopole = 2.7 and 4.6°, respectively), which is significantly different than the previous pole (13.0°N, 176.0°E, k = 30, dp = 6.0°, dm = 13.0°) reported by Vincenz and Yaskawa (1968). Halls and Palmer (1981) argued that the pole of Vincenz and Yaskawa (1968) was secondary, the result of a chemical remanent magnetization overprint. This study was undertaken to investigate the above discrepancy further and to obtain a paleomagnetic pole for a more complete sequence of lava flows within the Copper Harbor Conglomerate.

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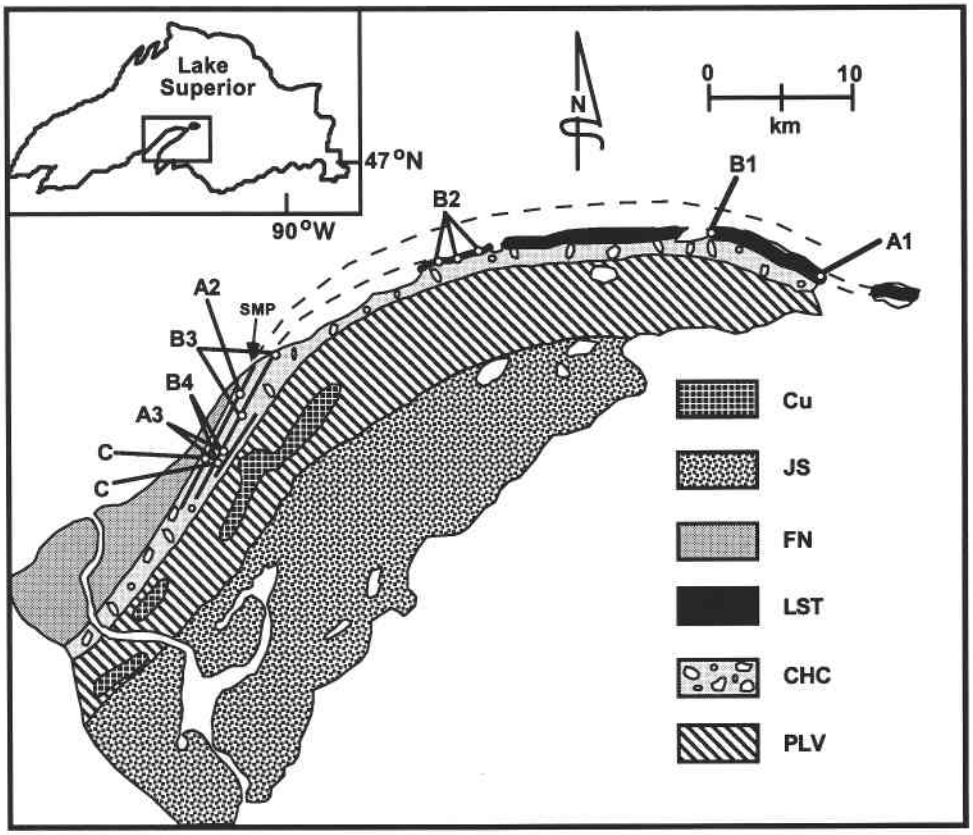


FIG. 1. Geological map of the Keweenaw Peninsula, Michigan. Shown are the sampling locations of paleomagnetic studies on the interbedded lava flows within the Copper Harbor Conglomerate. A1, sites north of High Rock Bay, this study; A2, Gratiot River sites, this study; A3, sites LT1 and LT2, this study; B1–B4, Halls and Palmer (1981); C, Vincenz and Yaskawa (1968) and Dubois (1962). Lithologies: Cu, areas of copper mineralization; JS, Jacobsville Sandstone; FN, Freda Sandstone and Nonesuch Shale; LST, Lake Shore Traps; CHC, Copper Harbor Conglomerate; PLV, Portage Lake Volcanics. Modified from Paces and Bornhorst (1985).

Geologic setting

The Copper Harbor Conglomerate is one of a number of rift-related geologic units that crop out along Michigan's Keweenaw Peninsula (Fig. 1). The unit is a reddish-brown, coarsely stratified pebble to boulder conglomerate with thin sandstone interbeds in its lower portion; higher in the section sandstone becomes the dominate facies. Conglomerate clasts are primarily rhyolitic, with approximately 10% being basaltic. The thickness of the Copper Harbor Conglomerate ranges from 200 to 2000 m (White and Wright 1960; Daniels 1982), thickening toward the Lake Superior basin. White and Wright (1960) and more recently Elmore (1984) have interpreted the Copper Harbor Conglomerate as coalesced alluvial fans and sand flats deposited by northward-flowing streams originating from hills along the southern margin of the Lake Superior basin. The Copper Harbor Conglomerate interfingers with and conformably overlies the Portage Lake Volcanics, a 2500–5200 m series of subaerial flood basalts dated at approximately 1.1 Ga (White 1960; Van Schmus et al. 1982; Davis and Paces 1990). The Nonesuch Shale interfingers with and conformably overlies the Copper Harbor Conglomerate, which in turn is comformably overlain by the Freda Sandstone.

Interbedded within the Copper Harbor Conglomerate is a succession of lava-flow units known collectively as the Lake Shore Traps (Lane 1911). These lava flows represent the youngest eruptive material associated with the 1.1 Ga Mid-continent rift system (Van Schmus et al. 1982). The Lake Shore Traps (LST) occur at three different stratigraphic levels (Fig. 2*a*) within the Copper Harbor Conglomerate (White et al. 1951) and display features typical of other Keweenawan subaerial flood basalts (Paces and Bornhorst 1985). The lowest lava flows of the LST are not exposed. They occur only in the western portion of the Keweenaw Peninsula and are inferred from magnetic surveys and drill-hole data (White et al. 1951). Only two lava flows, each less than 20 m thick, form the middle LST in the western portion of Keweenaw Peninsula (western half of Fig. 1). To the northeast, the middle LST thicken and the number of lava flows increases. At the eastern end of the Keweenaw Peninsula, the middle LST section consists of 31 individual lava flows totaling 594 m of thickness (Paces and Bornhorst 1985). The lava flows vary in thickness from 4.4 to 42.4 m (Fig. 2*b*) and are subdivided into two distinct stratigraphic packages by a 27 m thick interflow conglomerate layer. For the purposes of this paper, the 10 lava flows below and the 21 lava flows above the conglomerate shall be referred to as the *lower* and *upper* lava flows of the middle LST, respectively. The middle LST represent a stratigraphic succession of tholeiitic andesites, tholeiitic basaltic andesites, and iron-rich olivine tholeiites (Paces and Bornhorst 1985). A U–Pb date from a single lava flow from the middle LST yields an age of 1087.2 ± 1.6 Ma (Davis and Paces 1990). The outer LST is the uppermost lava-flow unit in the Copper Harbor Conglomerate. Southwest of the Seven Mile Point (SMP in Fig. 1), the outer LST section consists of 8–14

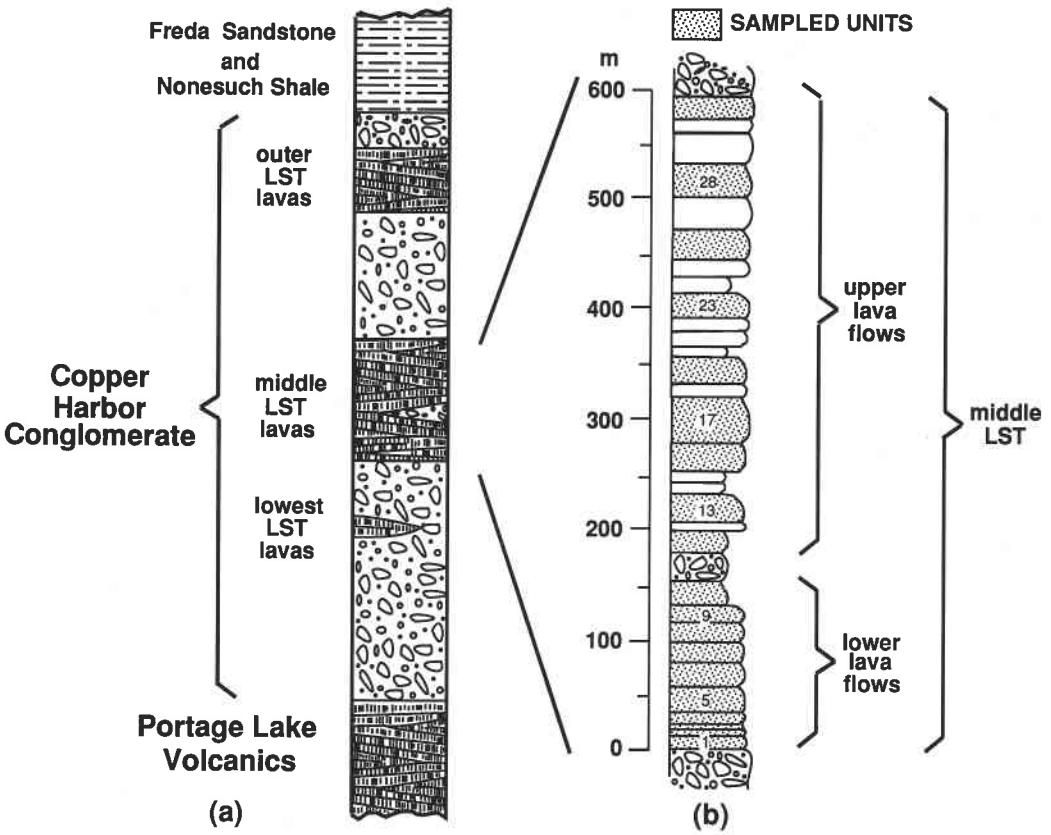


FIG. 2. (*a*) A schematic stratigraphic column of rock units in the western portion of the Keweenaw Peninsula showing the distribution of lava flow units (LST) within the Copper Harbor Conglomerate. (*b*) A detailed stratigraphic column of the middle LST at the eastern end of the Keweenaw Peninsula modified from Paces and Bornhorst (1985). Sampled lava flows are shown in shading. Numbers correspond to sampling sites listed in Table 2.

andesite lava flows and is approximately 150 m thick (White et al. 1951; J.B. Paces, personal communication, 1991); individual lava flows vary from 7 to 14 m. Northeast of Seven Mile Point, these lava flows manifest themselves only as magnetic anomalies in Lake Superior.

Previous studies

Dubois (1962), Vincenz and Yasakawa (1968), and Halls and Palmer (1981) have all published paleomagnetic data on the LST. In addition, Elmore and Van der Voo (1982) have studied the crossbedded sandstones of the Copper Harbor Conglomerate, and Palmer et al. (1981) have investigated the basaltic pebbles from the unit. Studies by Dubois (1962) and Vincenz and Yaskawa (1968) were based on a limited number of samples because each was a subsidiary portion of a larger study on Keweenawan rocks. Sample locations were confined to the central portion of the Keweenaw Peninsula (see Fig. 1) where the underlying Portage Lake Volcanics has undergone extensive native copper mineralization and low-grade burial metamorphism (White 1971; Stoiber and Davidson 1959). The most recent study on the LST is that of Halls and Palmer (1981). They also collected samples from sites in the vicinity of the mineralized area, but also farther east and north where mineralization and metamorphism are less intense in the underlying Portage Lake Volcanics (Stoiber and Davidson 1959).

Dubois (1962) studied only three sites from the middle and outer LST. Demagnetization techniques were not used. The measured directions from 13 samples yielded a pole positioned at 29.1°N, 182.8°E (k = 16.2, radius of the 95% confidence circle about the calculated mean pole determined using site-sample virtual geomagnetic poles A_{95} = 10.6°). More modern methods were used in a later study of the same lava-flow units by Vincenz and Yaskawa (1968) in which four sites from the outer LST and two sites from the middle LST were analyzed using alternating-field and thermal demagnetization. Characteristic directions were assumed to be those which exhibited the least directional change during the demagnetization process. Both alternating-field and thermal methods were successful in removing a soft viscous component resulting in a general shallowing of the remanent vector. The LST pole reported by Vincenz and Yaskawa (1968) was located at 13.0°N, 176.0°E (k = 30, dp = 6.0°, dm = 13.0°).

The study by Halls and Palmer (1981) concentrated on samples collected from eight sites located exclusively in the lava flows of the middle LST. Site locations extended approximately 50 km along strike from near the central mineralized area shown in Fig. 1 to the relatively less metamorphosed northeastern portion of the Keweenaw Peninsula. Halls and Palmer (1981) submitted their samples to both thermal and alternating-field demagnetization techniques. Components of magnetization having linear trends were isolated by the use of principle component analysis (Kirschvink 1980).

Halls and Palmer (1981) found two types of remanence. Sites farthest east and north from the zone of intense copper mineralization in the Portage Lake Volcanics contained only a single component of magnetization after the removal of a soft

present-day viscous magnetization. Results from a baked-contact test near the northernmost sampling location (Palmer et al. 1981) indicated that this magnetization was acquired during initial cooling of the flows and therefore was a primary magnetization. Sites collected nearer the mineralized area exhibited multicomponent behavior during both alternating-field and thermal demagnetization. When the primary or characteristic direction (declination $D = 299.0°$, inclination $I = 37.9°$) associated with the northernmost flows was isolated in these sites, it was found to be intermediate between a steeper, low-coercivity present-day viscous component and two shallower high-coercivity components ($D = 287.4°$, $I = 19.5°$, and $D = 284.7°$, $I = 9.3°$). The mean directions of the higher coercivity components are similar to (*i*) a pre-tilting partial chemical remagnetization ($D = 278.9°$, $I = 14.7°$) isolated in pebbles from within the Copper Harbor Conglomerate (Palmer et al. 1981); (*ii*) the primary magnetization direction ($D = 279.8°$, $I = 9.8°$) of the Nonesuch Shale as reported by Henry et al. (1977); (*iii*) the mean LST direction ($D = 282.0°$, $I = 12.0°$) reported by Vincenz and Yaskawa (1968); and (*iv*) the detrital remanence ($D = 297.0°$, $I = -7.6°$) and the chemical remanence ($D = 286.5°$, $I = 0.0°$) isolated by Elmore and Van der Voo (1982) from crossbedded sands of the upper Copper Harbor Conglomerate.

Based on (*i*), (*ii*), and (*iii*) above, Halls and Palmer (1981) concluded that the primary direction obtained by Vincenz and Yaskawa (1968) was thus secondary. The secondary direction was considered to be a chemical remanence that was acquired during the mineralization and burial metamorphism of the underlying Portage Lake Volcanics.

Paleomagnetic data

Sampling methods

A total of 236 oriented core samples were collected from 30 paleomagnetic sites (lava flows) within the Copper Harbor Conglomerate (Fig. 1). Twenty-two sites were collected from the middle LST; 20 were located along the shoreline north of High Rock Bay (A1 in Fig. 1) at the eastern end of the Keweenaw Peninsula (47.43°N, 87.71°W) and 2 were located in exposures southwest of Seven Mile Point (47.30°N, 88.45°W) in the western portion of the Peninsula (A3 in Fig. 1) and are equivalent to sites 5 and 6 (B4 in Fig. 1) of Halls and Palmer (1981). The remaining 8 sites sampled the outer LST along the Gratiot River (A2 in Fig. 1) near the western border of section 13, T57N, R33W of the Ahmeek Quadrangle (47.34°N, 88.45°W). These outer LST sites are approximately 3 miles (1 mile = 1.609 km) northeast of the previous sampling locations of Dubois (1962) and Vincenz and Yaskawa (1968).

Laboratory procedure

At least two pilot samples from each site were subjected to progressive alternating-field demagnetization (8–12 individual steps) to peak levels of 80–100 mT. One or two additional samples were subjected to stepwise thermal demagnetization using 7–12 heating steps per sample to temperatures of 670–690°C. When possible, alternating-field and thermal demagnetization studies were carried out on sister samples from the same core, and in some instances, thermal demagnetization was continued on samples previously subjected to alternating-field treatment to further reduce the sample's intensity to less than 5% of its original value. For sites located in lava flows of the outer LST, every sample was progressively demagnetized using either alternating-field or thermal or a combination of both. Measurement of weak field susceptibility before and after each heating and cooling cycle showed a lack of significant change during heating which indicated that no new material was created during thermal demagnetization.

The results of demagnetization experiments from each pilot sample were plotted on orthogonal vector endpoint diagrams using the method of Zijderveld (1967). Demagnetization levels that produced a linear decay of the vector end points to the origin in the pilot samples were then used on all remaining samples for that site. Characteristic remanent directions (ChRMs) were determined for all samples using principal component analysis (Kirschvink 1980). In general, five to eight demagnetization steps describing a linear demagnetization path to the origin were used to determine a sample's ChRM; less than 5% of the ChRMs were determined using three or four steps. Where the demagnetization data failed to describe a linear trend, a great circle was fit to the data. The resulting ChRMs were then used to calculate a site's mean direction using either the statistical methods of Fisher (1953) or those of McFadden and McEhlinny (1988). If two or more samples were cut from an individual core, their ChRMs were first averaged before calculating site means.

To characterize the magnetic mineralogy of the LST, representative samples were chosen for thermomagnetic, hysteresis, and petrographic analysis. Curie temperatures were determined using the graphical techniques of Gromme et al. (1969). Hysteresis data were compared with the results of Day et al. (1977) to characterize the domain state.

Laboratory results

Samples from the middle LST had initial intensities before demagnetization ranging between 1.10 and 72.0 A/m, with a geometric mean of 3.73 A/m for the lower lava flows, and the upper lava flows ranged in intensity between 0.46 and 89.3 A/m, with a geometric mean of 2.47 A/m. Samples from the middle LST south of Seven Mile Point (sites LT1 and LT2) had intensities between 1.7 and 7.8 A/m, with a geometric mean of 2.7 A/m. Outer LST samples had intensities from 0.30 to 59 A/m, with a geometric mean of 1.91 A/m.

Vector end-point diagrams illustrating the behavior of middle and outer LST samples during alternating-field and thermal demagnetization are shown in Figs. 3–5. All data are shown after correction for structural tilt. Demagnetization behavior was remarkably straightforward. Initially, a component of magnetization most often directed to the north and down before structural correction was removed. Upon continued demagnetization, a well-defined univectorial decay of the remanent vector to the origin results. The low-coercivity (blocking temperature) component is interpreted to be a present-day viscous overprint due to its similarity to the present Earth's field direction for this region. The magnitude of the overprint is highly variable being nonexistent in some samples (Fig. 5*b*) to comprising a considerable portion of the natural remanent magnetization (NRM) vector in others (Fig. 4*b*). In addition to this present-day overprint, other easily removed secondary overprints were present (Fig. 5*a*). The directions removed were quite variable and are probably the result of the viscous component being acquired in the laboratory. A single component of magnetization was isolated most often from about 40 mT onwards using alternating-field techniques and 400°C onwards using thermal techniques. Alternating-field and thermal demagnetization of sister samples from a single

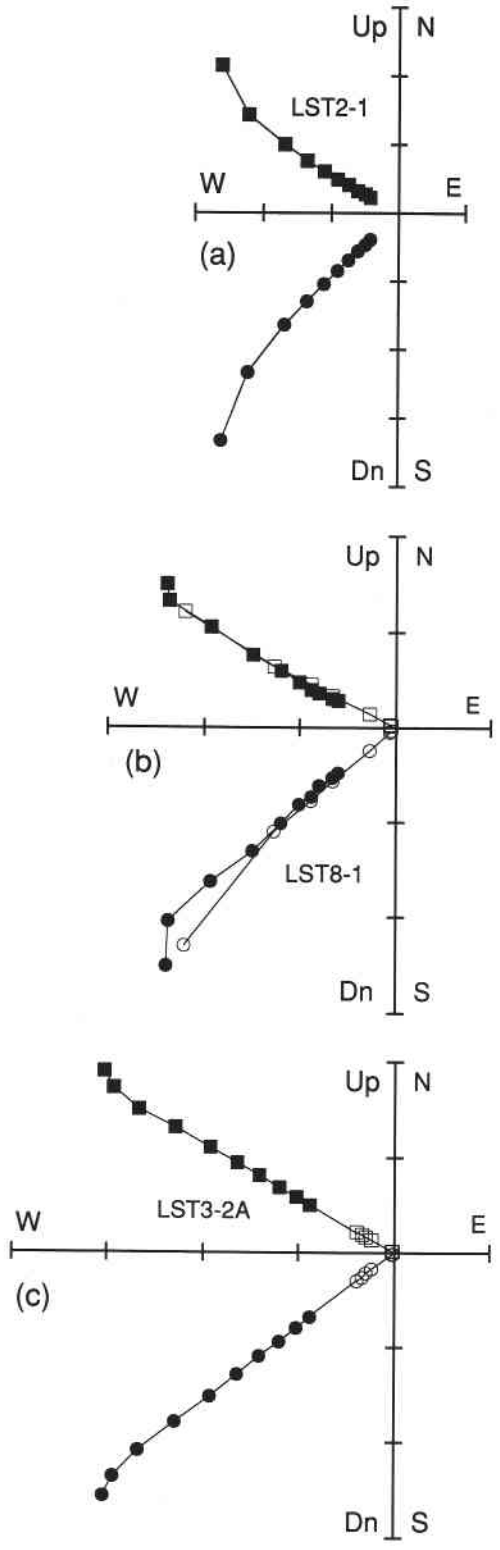


FIG. 3. Vector end-point diagrams for samples from the lower lava flows of the middle LST, illustrating their behavior to alternating-field (closed symbols) and thermal (open symbols) demagnetization. Squares are projections onto the horizontal plane; circles are projections onto the east–west vertical plane. Demagnetization steps are as follows: (*a*) 0, 10, 20, 30, 40, 50, 60, 70, 80, and 90 mT; (*b*) 0, 10, 20, 30, 40, 50, 60, 70, 80, and 90 mT, and 20, 530, 560, 590, 610, 640, and 670°C; (*c*) 0, 10, 20, 30, 40, 50, 60, 70, 80, and 90 mT, and 530, 560, 590, 610, 640, and 670°C. Horizontal and vertical axes divisions are 1.0 A/m. Dn, down.

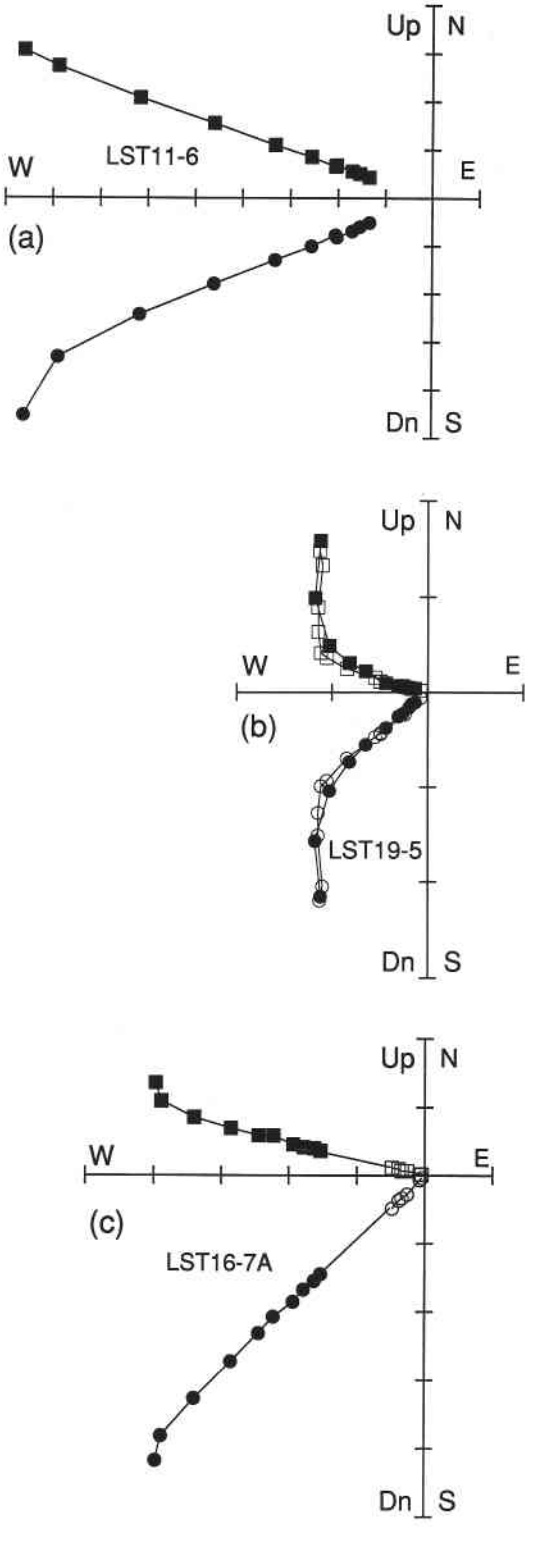


FIG. 4. Vector end-point diagrams of representative samples from the upper lava flows of the middle LST, illustrating their behavior to alternating-field (closed symbols) and thermal (open symbols) demagnetization. Squares are projections onto the horizontal plane; circles are projections onto the east–west vertical plane. Demagnetization steps are as follows: (*a*) 0, 10, 20, 30, 40, 50, 60, 70, and 80 mT; (*b*) 0, 5, 10, 15, 20, 30, 40, 50, 60, 70, and 80 mT, and 20, 100, 200, 300, 350, 400, 450, 500, 530, 550, 600, and 630°C; (*c*) 0, 10, 20, 30, 40, 50, 60, 70, 80, and 90 mT, and 530, 560, 590, 610, 640, and 670°C. Horizontal and vertical axes divisions are 0.1 A/m in (*a*), 0.5 A/m in (*b*), and 0.5 A/m in (*c*).

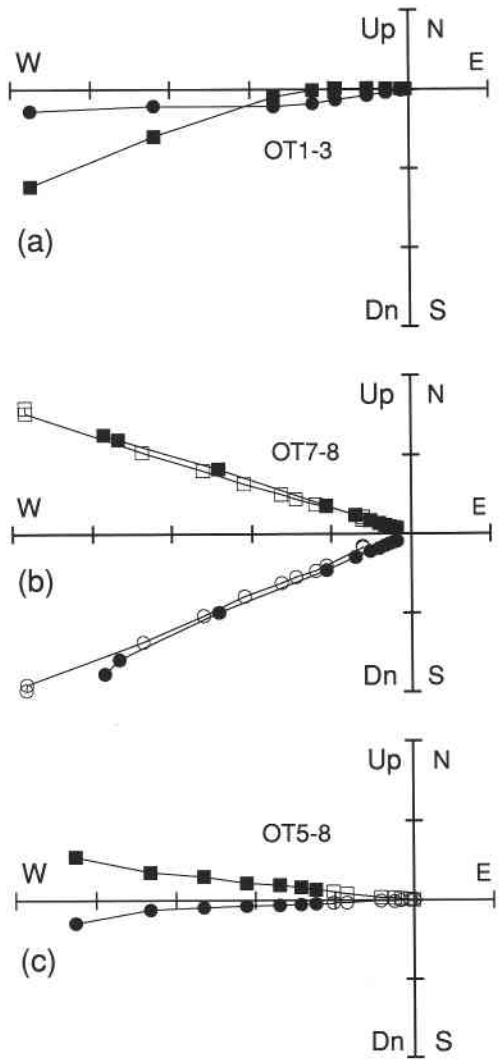


FIG. 5. Vector end-point diagrams of samples from the outer LST lava flows, illustrating their behavior to alternating-field (closed symbols) and thermal (open symbols) demagnetization. Squares are projections onto the horizontal plane; circles are projections onto the east–west vertical plane. Demagnetization steps are as follows: (*a*) 0, 5, 10, 15, 20, 30, 40, 60, and 80 mT; (*b*) 0, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100 mT, and 20, 100, 300, 350, 400, 450, 500, 530, 550, 600, and 630°C; (*c*) 0, 30, 40, 50, 60, 70, and 80 mT, and 530, 560, 590, 610, 640, and 670°C. Horizontal and vertical axes divisions are 1.0 A/m in (*a*), 10 A/m in (*b*), and 1.0 A/m in (*c*).

core (Figs. 3*b*, 4*b*, and 5*b*) isolate the same component of magnetization while thermal demagnetization of those samples with a significant amount of their original remanence remaining after alternating-field demagnetization continued the same linear trend to the origin (Figs. 3*c*, 4*c*, and 5*c*). Median demagnetizing fields averaged between 25 and 45 mT, with anywhere from less than 1 to 40% of the original NRM intensity remaining at peak inductions of 80–100 mT. Thermal demagnetization generally decreases the original intensity of a sample to less than 5% when heated above 610°C. In some instances, there is a noticeable decrease in sample intensity at heating temperatures between 560 and 590°C (Fig. 5*c*).

The ChRMs isolated by alternating-field methods are nearly identical (within 3°) to those unblocked by thermal methods. Linear segments to the origin, defining the characteristic direction for a sample, had an average maximum angular deviation angle of 0.9 and 1.1°, respectively, for the lower and upper lava flows of the middle LST, whereas sites LT1 and LT2 averaged 1.1°. The average angle of the outer LST sites was 1.5°. The maximum angular deviation angle is a quantitative measure of the precision with which a direction of magnetization has been determined from linear demagnetization path using the principal component method. Seventy-five percent of the outer LST samples had maximum angular deviations less than or equal to 2°, whereas 95% of the middle LST samples were less than or equal to 2°.

Four samples from site OT7 and one sample from site OT8 from the outer LST proved to be the exception to the demagnetization behavior discussed above. These samples displayed vector end-point diagrams with curved trajectories defining a great circle path on a stereonet. They had unusually high intensities (>40 A/m) and large differences between sun and magnetic compass orientations. Sample intensities decreased rapidly with alternating-field demagnetization, with less than 50% of the original intensity remaining by 7.5 mT. Given the above observations we surmised that these samples acquired a lightning-induced magnetization that could not be entirely removed by either alternating-field or thermal demagnetization.

The demagnetization behavior of our samples from the middle LST collected at the eastern end of the Keweenaw Peninsula is similar to that reported by Halls and Palmer (1981) for sites collected in the middle LST in locations farther west along the peninsula (B1 and B2 in Fig. 1). Halls and Palmer (1981) did not sample the flows of the outer LST.

Curie temperature and hysteresis results from the middle and outer LST are given in Table 1. Thermomagnetic curves (Fig. 6) exhibit reversible heating and cooling cycles. In all cases only a solitary carrier of magnetic remanence was seen. Curie temperatures were near 580°C and indicate that the primary carrier of remanence is a titanium-poor magnetite. Previous investigations on the outer LST by Vincenz and Yaskawa (1968) reported a Curie temperature of 565°C. The hysteresis parameters J_{rs}/J_s and H_{rc}/H_c (defined in Table 1) indicate that the magnetite grains are pseudo-single domain. Sample LST1-8 (Fig. 6*b*) proved to be the exception, exhibiting a Curie temperature of 505°C with hysteresis parameters suggesting the carrier of remanence is multidomain.

Analysis of specimens from the middle LST under reflected light indicated that the LST have undergone various degrees of high-temperature oxidation during initial cooling. Samples show varying degrees of ilmenite exsolution along the 111 crystallographic planes of equant, subhedral titanomagnetite grains ranging from 0.01 to 0.2 mm in size. Sample LST1-8, however, showed little if any exsolution lamellae within very homogeneous titanomagnetite grains. Signs of varying degrees of hematization were seen in the form of filling along fractures, or replacement at the border of titanomagnetite grains.

Iron oxides in the outer LST ranged in size from 0.02 to 0.3 mm and were distributed evenly throughout the sample. Again the titanomagnetites host ilmenite lamellae along the 111 crystallographic planes. Hematite was present as both mottling in titanomagnetite grains and as cavity filling along minute fractures. In general, petrographic observations were similar to those reported by Vincenz and Yaskawa (1968).

The petrographic and demagnetization results indicate that a titanium-poor magnetite is the predominant carrier of remanence in the LST, although hematite contributes. Since the

TABLE 1. Curie temperatures and hysteresis results

Sample No.	J_{rs} (A · m²/kg)	J_s (A · m²/kg)	H_{rc} ($\times 10^{-4}$ T)	H_c ($\times 10^{-4}$ T)	J_{rs}/J_s	H_{rs}/H_c	Domain	T_c (°C)
LST1-8	0.072	1.05	327	55	0.069	5.95	MD	505
LST7-8	0.125	1.26	405	110	0.100	3.68	PSD	572
LST17-5	0.138	1.03	543	208	0.134	2.61	PSD	570
LST26-1	0.111	0.61	380	170	0.182	2.23	PSD	580
LST31-8	0.145	1.24	304	113	0.117	2.95	PSD	582
OT5-2	0.538	1.77	375	165	0.304	2.27	PSD	594
OT7-8	0.287	0.87	523	220	0.330	2.38	PSD	585

NOTES: J_s, saturation magnetization; J_{rs}, remanent saturation magnetization; H_c, coercivity; H_{rc}, coercivity of remanence; PSD and MD, pseudo-single domain and multidomain, respectively (Day et al. 1977); T_c, Curie temperature.

direction that resides in the hematite component is nearly identical to that in the magnetite component, the hematite is probably deuteric and formed during the initial cooling of the LST lava flows.

Discussion

Paleomagnetic directions

Table 2 summarizes the paleomagnetic data obtained from the LST after alternating-field and (or) thermal demagnetization. Although individual sample directions are well defined, three samples, one each from sites OT2, OT4, and OT5, had ChRMs that were very distinct (>30°) from the rest of the ChRMs from that site. To avoid any bias these outliers might introduce in our calculations, these samples were eliminated from their respective sites before final statistical analysis. As seen in Fig. 7, the site-mean directions tend to cluster in three distinct groups, with several sites displaying intermediate directions. The three groups correspond to directions obtained from the lower lava flows of the middle LST, the upper lava flows of the middle LST, and the outer LST. Intermediate directions come from site LST11 from the upper lava flows of the middle LST, and sites LT1 and LT2 from the middle LST in the western portion of the Keweenaw Peninsula southwest of Seven Mile Point (A3 in Fig. 1).

The mean direction (D = 300.0°, I = 34.9°, k = 425.6, radius of the 95% confidence circle about the calculated mean direction determined using site mean directions α_{95} = 2.3°) determined for the lower lava flows of the middle LST is statistically indistinguishable from the direction (D = 299.0°, I = 37.9°, k = 221.8, α_{95} = 4.1°) reported by Halls and Palmer (1981) for the same lavas sampled at several locations to the west. A baked contact test reported by Palmer et al. (1981) substantiated that the remanence of the middle LST lavas was acquired during initial cooling of the flows. The mean direction (D = 279.3°, I = 38.7°, k = 97.6, α_{95} = 4.9°) for the upper flows of the middle LST, however, has a statistically different direction. Although a baked contact test was not obtained from the conglomerate that directly underlies these lava flows, there is nothing to indicate that the ChRMs isolated in the upper lava flows of the middle LST were not acquired at the time of initial cooling.

The low inclination direction (D = 277.8°, I = 5.2°, k = 344.7, α_{95} = 3.0°) determined from the eight outer LST sites as well as sites LT1 and LT2 requires closer scrutiny. This direction is similar to inclinations isolated in sediments from the Copper Harbor Conglomerate, several of which have been interpreted as chemical remagnetizations (Palmer et al. 1981; Elmore and Van der Voo 1982). Since the outer LST sites as well as sites LT1 and LT2 are located directly upsection from the most mineralized portion of the underlying Portage Lake Volcanics (see Fig. 1), the low inclinations may be an effect of chemical−thermal overprints associated with this mineralization as suggested by Halls and Palmer (1981) in their study on the middle LST. However, this is probably not the case. The low inclination directions observed in these lava flows reside both in magnetite and hematite and are parallel (Fig. 5*c*). Beske-Diehl and Li (1993) in their study of Icelandic lava flows have shown that magnetite and hematite retain their primary directions of magnetization to the bottom of the laumonite zeolite zone. Since the lava flows of the LST have only undergone laumonite zeolite grade metamorphism (Stoiber and Davidson 1959; Bornhorst et al. 1983), we would argue that chemical overprinting is not the cause of the low inclination directions observed in the outer LST sites as well as sites

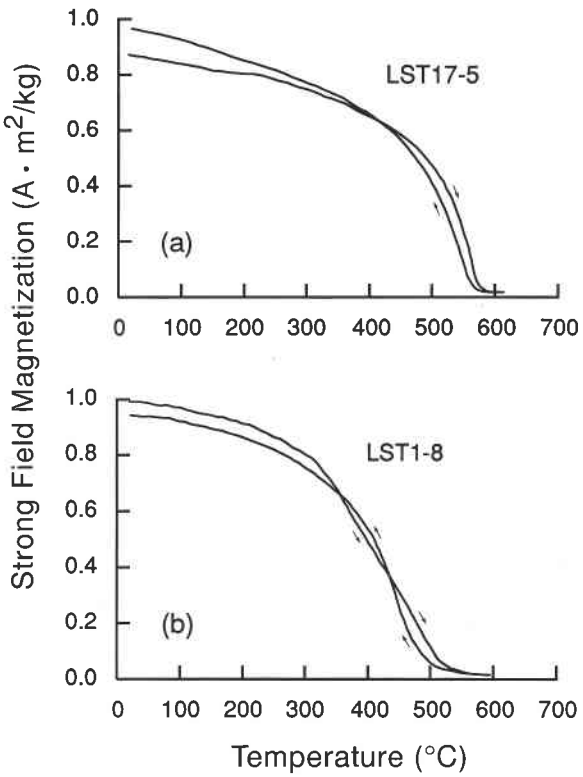


FIG. 6. Representative thermomagnetic curves for the Lake Shore Traps. Samples were heated and cooled in vacuum in a field of 0.2 T.

TABLE 2. Summary of paleomagnetic data from the interbedded lava flows within the Copper Harbor Conglomerate, Keweenaw Peninsula, Michigan

Site	N	D (°)	I (°)	α_{95} (°)	k	Plat. (°)	Plong. (°)
Middle LST, eastern sites							
Lower lava flows							
LST1	9	301.3	38.7	3.8	184.7	36.9	175.8
LST2	7	302.5	37.6	4.8	162.8	37.1	173.2
LST3	7	302.5	36.7	2.4	620.2	36.7	172.6
LST4	8	300.1	36.8	2.1	681.1	35.2	174.6
LST5	7	302.0	34.8	2.4	630.0	35.5	171.9
LST6	9	300.4	35.1	3.4	228.2	34.6	173.4
LST7	10	295.5	35.6	2.0	581.8	31.5	177.6
LST8	8	299.7	35.2	3.9	207.3	34.1	174.0
LST9	8	295.8	27.6	3.9	206.4	28.1	173.2
LST10	7	300.7	30.7	2.1	863.3	32.7	170.8
Upper lava flows							
LST11	7	289.9	20.9	3.1	379.3	21.4	174.9
LST13	8	285.3	42.4	4.5	155.4	27.9	189.1
LST16	9	278.7	42.6	2.7	358.8	23.6	193.7
LST17	8	278.9	43.9	3.3	292.0	24.4	194.4
LST19	8	282.5	39.9	3.1	331.1	24.8	189.5
LST23	8	276.9	41.9	2.0	786.0	22.1	194.5
LST26	8	271.9	41.9	5.5	103.5	18.8	197.9
LST28	8	276.4	32.4	3.9	207.2	17.1	189.9
LST29	9	276.2	39.5	2.1	592.9	20.4	193.6
LST31	8	274.2	40.2	3.9	208.3	19.4	195.4
Middle LST, western sites							
LT1	7	295.4	12.9	2.4	656.2	21.9	166.8
LT2	7	281.8	22.3	3.4	321.6	16.5	180.9
Outer LST							
OT1	8	281.3	8.9	6.1	83.6	10.9	176.3
OT2	7	275.4	3.9	5.8	109.0	5.1	178.9
OT3	7	275.3	1.5	4.2	206.6	4.1	178.2
OT4	6	278.4	3.6	6.4	109.7	7.0	176.6
OT5	7	276.5	4.7	3.0	394.1	6.1	178.3
OT6	7	275.9	2.9	4.5	182.8	5.1	178.2
OT7	9	283.9	10.2	5.8	83.6	13.2	174.8
OT8	7	275.6	5.8	2.1	863.3	5.9	179.4
Mean direction and pole							
	30	286.1	27.5	5.9	20.9		
	30			4.5	35.0	22.2	180.8

NOTES: N, number of samples (sites); D and I, mean declination and inclination, respectively; α_{95} and k, Fisher (1953) statistical parameters; Plat. and Plong., north latitude and east longitude, respectively, of virtual geomagnetic pole position. All data are structurally corrected. Dip azimuth and angles used to restore the lava flows to the horizontal are as follows: 22 and 24° for the middle LST, eastern sites; 298 and 29° for the middle LST, western sites; and 295 and 25° for the outer LST sites.

LT1 and LT2. Additionally, given that magnetite and hematite have different time–temperature relationships (Pullaiah et al. 1975), it is highly unlikely that a chemical–thermal overprint would produce similar directions in both magnetite and hematite given the low degree of burial metamorphism.

Halls and Palmer (1981), in their study on the middle LST, considered that the segment of the demagnetization path whose direction was similar to that from the positive baked contact test of Palmer et al. (1981) represented the primary direction for these lava flows. Therefore, the shallow higher coercivity component, which did decay to the origin and was found predominantly in their westerly sites (e.g., sites 5, 6, 7, and 11; locations B3 and B4 in Fig. 1), was considered secondary because it was not similar to that from the baked contact test. This is valid only if the lava flows sampled in the western portion of the Keweenaw Peninsula are indeed the time equivalents of those sampled to the east. Our study indicates that there is considerable directional change between the lower and upper lava flows of the middle LST. Furthermore, site LST11, the lowest unit in the upper lava flows of the middle LST, has a mean direction (D = 289.9°, I = 20.9°, α_{95} = 3.1°) very similar to that of site LT2 (D = 281.8°, I = 22.3°, α_{95} = 3.4°), which is the same lava flow (site 6) whose direction (D = 282.8°, I = 22.8°, α_{95} = 4.3°) Halls and Palmer (1981) interpreted as a chemical remanent overprint. The middle LST thins from a maximum of 31 lavas at the eastern end

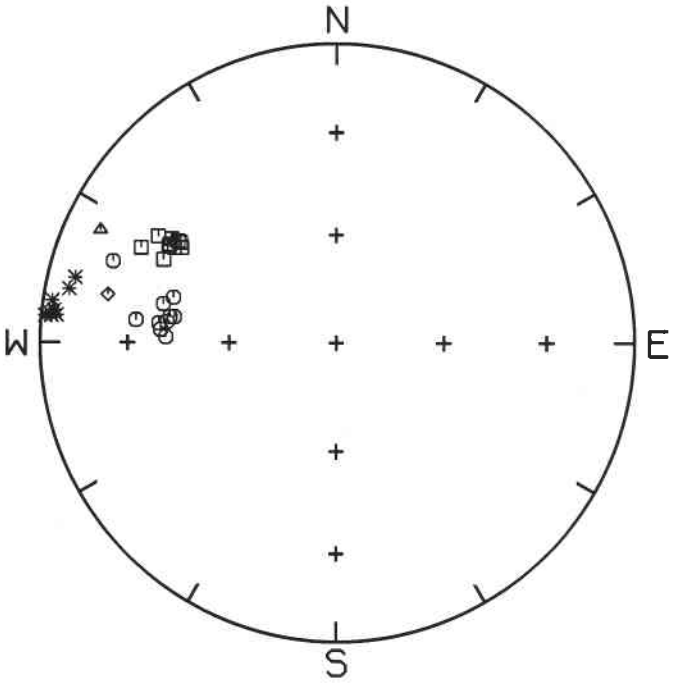


FIG. 7. Site-mean directions from the lava flows of the LST (all lower hemisphere). □, lower lava flows of the middle LST; ○, upper lava flows of the middle LST; *, outer LST lava flows; △, LT1; ◇, LT2. LT1 and LT2 are both middle LST sites from the western portion of the Keweenaw Peninsula (location A3 in Fig. 1).

of the Keweenaw Peninsula (Paces and Bornhorst 1985) to two near the area of extensive native copper mineralization in the underlying Portage Lake Volcanics (White 1951), where we and Halls and Palmer (1981) have sampled. The possibility exists that the flows which Halls and Palmer sampled in the eastern portion of the Keweenaw Peninsula are not time equivalents of the lava flows sampled in the western portion, thus rendering their identification of primary and secondary components questionable. In fact, what Halls and Palmer considered primary in specimens from their western sites (i.e., 5, 6, and 7) may merely represent a transition from a present-day overprint to a shallow inclination direction, as this primary direction is only isolated when a present-day component is present. This interpretation is also suggested by Halls and Palmer (1981) in Appendix 1 of their paper. Therefore, we suggest that the shallow inclination direction determined from the flows of the outer LST and from sites LT1 and LT2 is not a chemical remanent magnetization overprint but in all probability a primary magnetization acquired at the time of cooling when the geomagnetic field had a low inclination direction.

Lastly, as noted earlier, the site-mean directions determined from the lower and upper lava flows of the middle LST at the eastern end of the Keweenaw Peninsula, as well as those from the outer LST lava flows, tend to form tight clusters that are very distinct. Between-site dispersion values for these groups, as given by the angular standard deviation (θ_{63}), are rather low: 3.9° for the lower lava flows of the middle LST; 8.2° for the upper lava flows of the middle LST, which drops to 4.6° if site LST11 is eliminated from the calculation; and 4.4° for the outer LST lavas. These values have not been corrected for within-site dispersion and therefore are maximum estimates of the between-site scatter. Between-site dispersion is not only a function of paleolatitude of the sampling sites (Creer 1962) but more importantly for this study is also a measure of how efficiently these sites average out the dominant periodicities of the secular variations of the ancient geomagnetic field. Thus, a comparison of our between-site dispersions for each group of flows to some "expected" dispersion value should indicate how well each group has averaged out secular variation. Halls and Pesonen's (1982) study of Keweenawan igneous units of normal polarity gives an average between-site dispersion of 11.1° (Table 3, p. 188), with lower and upper 95% confidence bounds of 9.0 and 14.6°, respectively. Using the method of Cox (1969), only the sites located in the upper flows of the middle LST have a between-site dispersion that is statistically similar to other Keweenawan rock units from the Lake Superior region. If site LST11 is excluded, then all three lava flow units have dispersions that are statistically lower than the expected 11.1°. However, since we sampled only 10 of the 21 lava flows forming the upper middle LST, our sampling scheme may be biasing our results. Nevertheless, we suggest that the low between-site dispersion results because these lava flow units were rapidly extruded, not allowing paleosecular variation to be adequately averaged. This is consistent with geological evidence (i.e., superposed flows and no intercalated paleosols) indicating these lava flows were emplaced rather rapidly. Therefore, a paleomagnetic pole from a single lava flow unit within the Copper Harbor Conglomerate will in all probability not be a time-averaged value of the late Precambrian geomagnetic field (i.e., a true paleomagnetic pole) but a spot reading of the field (i.e., a virtual geomagnetic pole).

Paleomagnetic poles

Figure 8 shows the distribution of site-mean virtual geomagnetic poles for all 30 LST sites. Also shown is the mean pole for the Copper Harbor Conglomerate determined from the LST virtual geomagnetic poles and its circle of 95% confidence. Like the site-mean directions (Fig. 7), virtual geomagnetic poles from the same LST lava flow unit cluster together but show slightly more dispersion, a result of mapping directions into poles. In calculating the position of the paleomagnetic pole for the Copper Harbor Conglomerate, we have chosen to give unit weight to each LST site-mean virtual geomagnetic pole. We do so realizing that not all the virtual geomagnetic poles from the LST are independent spot readings of the geomagnetic field. Since there is approximately the same number of virtual geomagnetic poles from each cluster, no one group or direction will dominate our calculation using this method. However, in doing so, we overestimate the confidence limits (i.e., A_{95}) on the mean pole, since A_{95} is dependent on N, the number to site-mean virtual geomagnetic poles. A more conservative approach in calculating the confidence limits about the mean pole would be to assume that only half the sites are independent (a figure not unrealistic in light of the distribution shown in Fig. 7), but the precision of the data set remains unchanged (i.e., $k = 35$). Using the approximation equation for A_{95} ($A_{95} = 140/\sqrt{kN}$), we calculate a A_{95} of 6.1° (instead of the original 4.5°). It is this value that is plotted in Fig. 8.

Figure 9 shows the distribution of paleomagnetic poles from the Portage Lake Volcanics, the Copper Harbor Conglomerate, the Nonesuch Shale, and the Freda Sandstone. The stratigraphic order of these units is well established. Also shown in Fig. 8 is the paleomagnetic pole for the Michipicoten Island Formation (Palmer and Davis 1987). The interpreted age of these rocks (~1087 Ma) is identical to the isotopic age of the LST. In general, paleomagnetic poles from rock units of

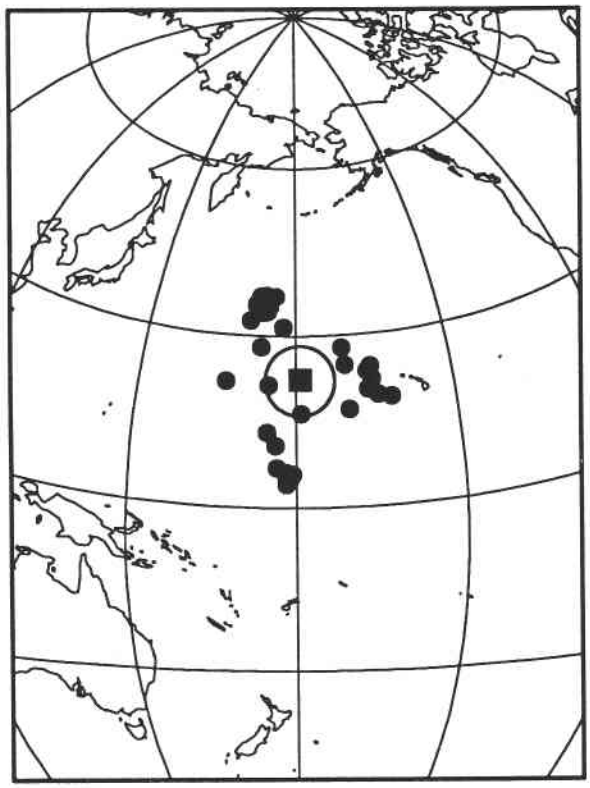


FIG. 8. An oblique equal-area projection showing the distribution of site-mean virtual geomagnetic poles (●) from the LST lava flows and the mean paleomagnetic pole (■) (22.2°N, 180.8°E) with its circle of 95% confidence (A_{95} = 6.1°).

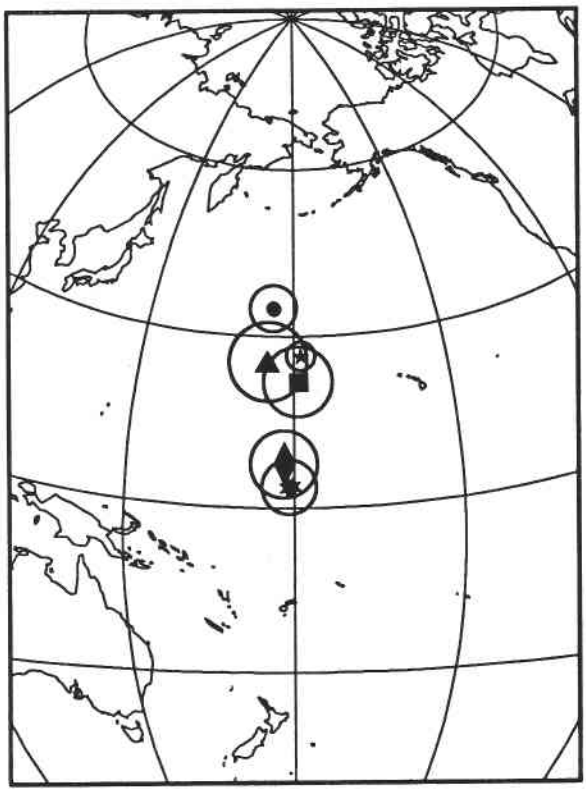


FIG. 9. Selected paleomagnetic poles from Keweenawan units from the Lake Superior basin. ●, middle LST lavas (Halls and Palmer 1981); ★, Portage Lake Volcanics (Halls and Pesonen 1982); ■, this study; ▲, Michipicoten Island Formation (Palmer and Davis 1987); ♦ and ✱, Nonesuch Shale and Freda Sandstone, respectively (recalculated from Henry et al. (1977) using site-mean virtual geomagnetic poles).

Keweenawan age from the Lake Superior region sweep out a path of apparent polar wander that youngs in age in a counterclockwise, north–south direction (Halls and Pesonen 1982). The position of our new Copper Harbor pole is significant in comparison to previously determined poles for the Copper Harbor Conglomerate from the LST and other Keweenawan-age rocks. The previous pole position calculated for the Copper Harbor Conglomerate by Halls and Palmer (1981) from the LST lies to the north of the Portage Lake Volcanics pole, a position that would make the Copper Harbor Conglomerate older than the Portage Lake Volcanics based on the systematics of Keweenawan APW. In addition, this pole is statistically distinct from the Michipicoten Island pole. Conversely, the Copper Harbor pole reported in this study is statistically similar to the Michipicoten Island pole and is positioned properly with respect to pole from the underlying (older) Portage Lake Volcanics and the poles from the overlying (younger) Nonesuch Shale and Freda Sandstone on the Keweenawan apparent polar wander path. By sampling only one stratigraphic unit of the LST within the Copper Harbor Conglomerate, Halls and Palmer (1981) and others (Dubois 1962; Vincenz and Yaskawa 1968) have failed to sample the full range of directions recorded by the LST.

The proximity of our Copper Harbor Conglomerate pole to the Portage Lake Volcanics pole is chronologically important. Previous paleomagnetic studies (Dubois 1962; Vincenz and Yaskawa 1968; Halls and Palmer 1968) have reported conflicting evidence as to the temporal relationship of the Copper Harbor Conglomerate to the underlying Portage Lake Volcanics and the overlying Nonesuch Shale and Freda Sandstone. The results of this study indicate that the Copper Harbor Conglomerate is more closely related in time to the Portage Lake Volcanics than to the Nonesuch Shale and Freda Sandstone. If the rate of apparent polar wander has remained constant during deposition of the Portage Lake Volcanics, the Copper Harbor Conglomerate, the Nonesuch Shale, and the Freda Sandstone, a reasonable assumption, then the transitional boundary between the Copper Harbor Conglomerate and the Nonesuch Shale represents a longer time period than the transitional boundary between the Copper Harbor Conglomerate and the Portage Lake Volcanics. The extended duration of the transition from Copper Harbor Conglomerate to Nonesuch Shale probably reflects the drastic change in sedimentation rates between these two units and the transformation of the rift from a rapidly subsiding volcanic basin to a more slowly subsiding lacustrine basin (Davis and Paces 1990; Cannon 1992). Recently, the United States Geological Survey revised the stratigraphic nomenclature within the Keweenawan Supergroup of northern Michigan (Cannon and Nicholson 1992). In doing so, Cannon and Nicholson (1992) reassigned the Copper Harbor Conglomerate from "middle Keweenawan" of old usage (e.g., White 1972) to the basal unit of the Oronto Group. Based on the above, we would support the arguments of White (1972) that the top and not the bottom of the Copper Harbor Conglomerate is a more appropriate boundary for the base of the Oronto Group.

Conclusions

(1) The middle LST lava flows sampled along the shoreline north of High Rock Bay at the eastern end of the Keweenaw Peninsula and southwest of Seven Mile Point and the lava flows of outer LST along the Gratiot River are magnetically stable, yield well-defined ChRMs, and are relatively unaffected by secondary overprinting. The carrier of remanence in these flows is for the most part a pseudo-single domain titanium-poor magnetite that has undergone some high-temperature oxidation.

(2) The site-mean directions and virtual geomagnetic poles determined from lower and upper lava flows of the middle LST and for the outer LST flows define three distinct clusters with some site means and virtual geomagnetic poles distributed between the clusters. Between-site dispersion for the three stratigraphic units sampled is much less than that which would be expected for Keweenawan-age rocks. This would indicate that most of the lava flows in each unit were extruded rapidly and therefore have not adequately sampled paleosecular variation. The angular separation between clusters of directions–poles is thus due to paleosecular variation, and not apparent polar wander, or chemical overprints. A mean pole–direction calculated from anyone of the three packages of flows would lead to an erroneous positioning of the pole. This explains why previous studies on the LST have produced such different results.

(3) The new Copper Harbor pole position determined from LST lava flows now lies in the appropriate chronological position between the Portage Lake Volcanics and the Nonesuch Shale on the Keweenawan apparent polar wander path. The proximity of this pole to the Portage Lake pole suggests that the Copper Harbor Conglomerate is more closely related in time to the underlying Portage Lake Volcanics than to the overlying Nonesuch Shale.

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